

# Result Reproduction

## 1 Goncalves et al., “Numerical solution of the time dependent Ginzburg-Landau equations for mixed (d + s)- wave superconductors”

We try to reproduce section VIII, the paper’s computational results. The parameters are  $\kappa_D = 2, \tau_1 = 8/3, \tau_3 = 16/3, \tau_4 = 2, \nu = -2, \eta_s = 2, \eta_v = 1$ . Note that these correspond to microscopically derived coefficients from other papers. The square film has dimensions  $8\xi_D \times 8\xi_D$ . The grid starts off with (everywhere)  $\psi_D = 1, \psi_S = 0, \mathbf{B} = 0$ . In the paper, they gradually sweep the field up to  $2.5H_{c2}$  and back down to 0. My computer only had the capability to step in intervals of  $0.5H_{c2}$ , which probably accounts for the differences in behavior below. Another possibility is that solving on a mesh rather than a grid causes differences, but I think that’s less likely.

One issue, among many other differences (due to my intervals), is that there is no vortex remaining when the system gets back to  $H = 0$ . However, the vortex count seems to roughly track otherwise.

I tried to reproduce these plots by going straight from  $H = 0$  to the desired values, but again hysteresis seems to cause behavioral differences (in mine there are 8 vortices rather than 6).

## 2 Xu, Ren, Ting, “Structures of single vortex and vortex lattice in a d-wave superconductor”

## 3 Lin et al., “Distinguishing between s+id and s+is pairing symmetries in multiband superconductors through spontaneous magnetization pattern induced by a defect”

## 4 Lei et al., “Field Driven Pairing State Phase Transition in $d_{x^2-y^2} + id_{xy}$ -Wave Superconductors”

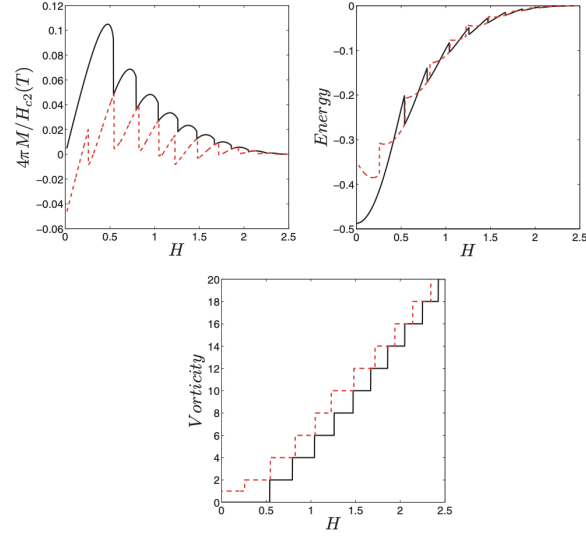
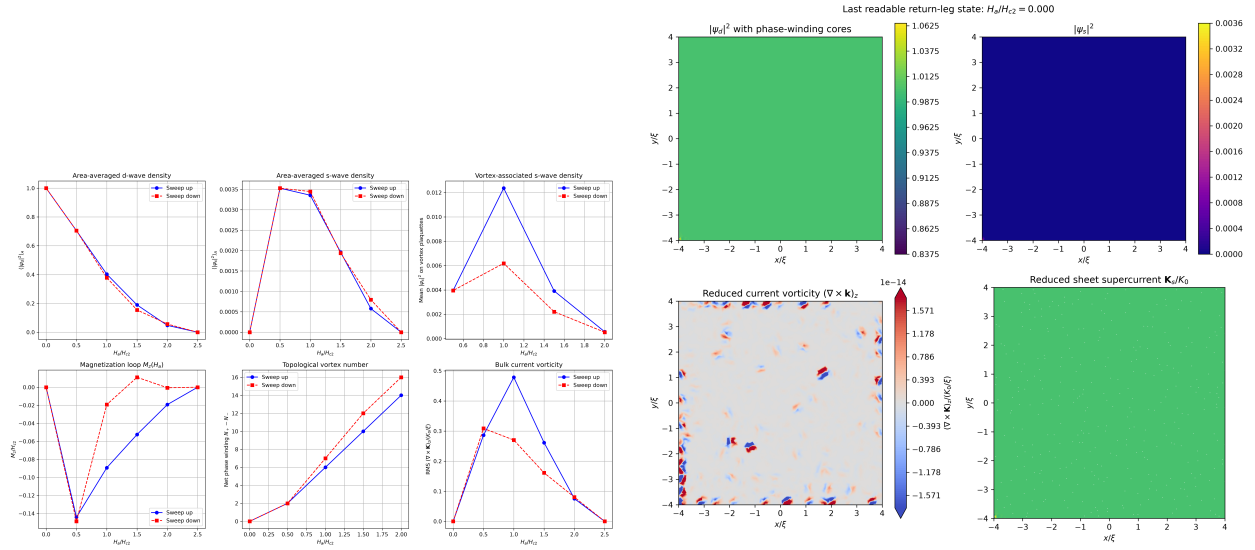


FIG. 2. The magnetization (upper-left) and the Gibbs free-energy (upper-right) as a function of the external applied field; for magnetic field sweeping up (black curve) and then down (red dashed curve). Different states are labeled by their corresponding vorticity  $L$  (lower panel).



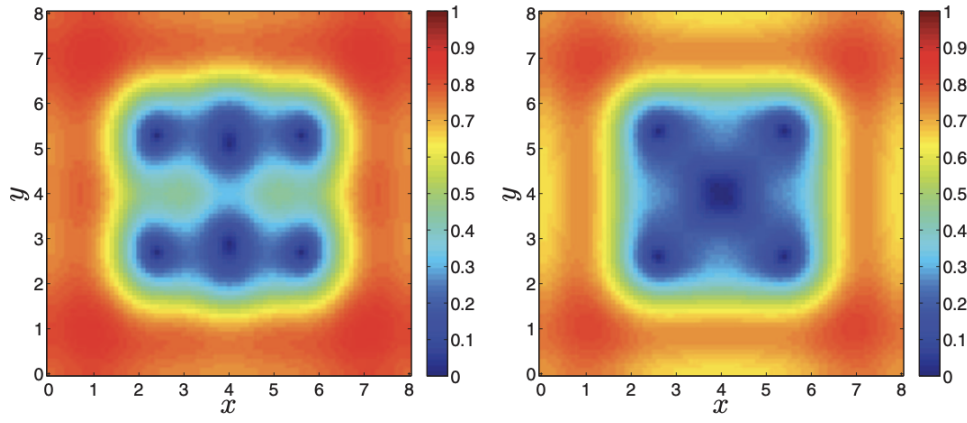


FIG. 6. Density plot of  $|\psi_d|$  for six vortices at  $H = 1.040$  (left) and  $H = 1.125$  (right).

